Assignment 8

November 15, 2018

Question 8

(a) "Read the data into SAS and display the data set"

We read the data into SAS as follows

```
filename myurl url " http://www.utsc.utoronto.ca/~butler/c32/heart-rates.csv";

proc import
    datafile=myurl
    dbms=csv
    out=mydata
    replace;
    getnames=yes;

proc print;
```

and get the following output (this is purposely cut down to only 10 rows):

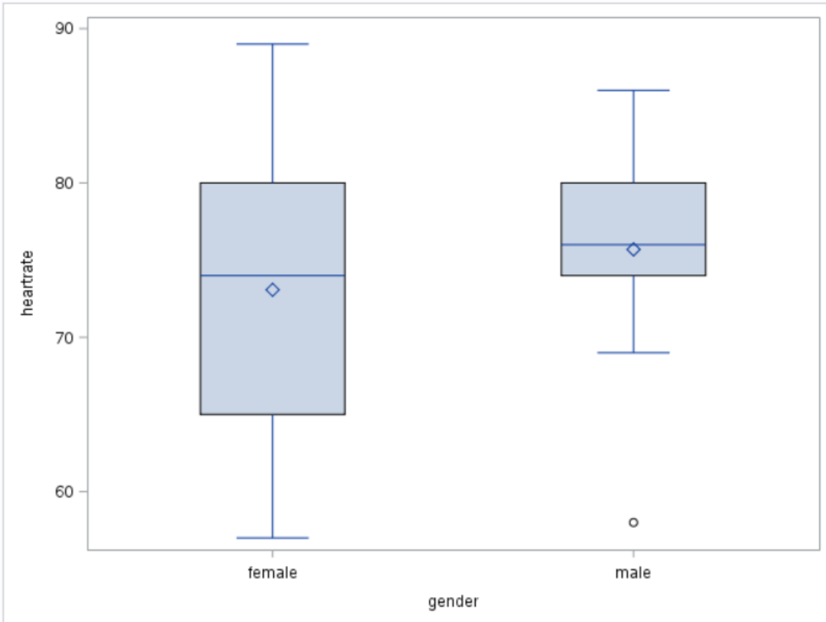
Obs	gender	heartrate
1	male	74
2	male	80
3	male	75
4	male	69
5	male	58
6	male	76
7	male	78
8	male	78
9	male	86
10	male	84

(b) "Make a suitable plot of the two variables"

From the output above, we see that there are two variables. One is "gender" which is a categorial variable, and the other is heartrate which is a quantitative variable. The code is found below:

```
proc sgplot;
    vbox heartrate / category=gender;
```

And the output is found below:



(c) "Run the most appropriate t-test to compare the mean heart rate for males and females. What do you conclude, in the context of the data?"

Though it is not suggested to conduct a t-test, let's conduct one anyway. Formally, we present our null and alternative hypothesis as follows:

$$H_0 : \mu_m = \mu_f$$
$$H_a : \mu_m \neq \mu_f$$

with an alpha of $\alpha = 0.05$ where μ_m is the mean heart rates for males and μ_f is the mean heart rates for females. Informally, our null hypothesis is that the mean heart rates between the two genders are the same, and the alternative hypothesis is that the mean heart rates between the two genders aren't the same.

Our code is found here:

```
proc ttest;
    var heartrate;
    class gender;
```

and the relative output from running the code above is found here:

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	24	-0.74	0.4652
Satterthwaite	Unequal	21.059	-0.74	0.4662

In both the Pooled t-test and the Satterthwaite t-test, the p-value is approximately 0.47. In this case, we fail to reject null hypothesis (since $0.47 > 0.05$) and conclude that the mean heart rates between genders are the same *under the assumption of normality*.

(d) "Would you trust the result of your test? Explain briefly why or why not. Refer back to any of your previous output as necessary"

I do not trust the result of my test. Judging by my boxplot in (b), there was an outlier found in the male's boxplot at the bottom. T-tests are supposed to assume normality, but having this disrupts normality (as it may push the mean down). Central limit theorem barely helps out for such a small sample size (as it is less than 30).

Question 9

A study is being done to test whether a population mean is 100. The population standard deviation is believed to be about 10, and the population is believed to be normal shape.

(a) "If a sample of size 35 is taken, and the population mean is actually 103, how likely is it that the null mean of 100 will be correctly rejected against a two-sided alternative?"

Since there is no categorizing value in our population, we can assume "onesamplemeans". Our code is found below:

```
proc power;
    onesamplemeans
    test=t
    mean=100
    nullmean=103
    stddev=10
    ntotal=35
    power=.;
```

And the output from the code above is as follows:

The POWER Procedure
One-Sample t Test for Mean

Fixed Scenario Elements	
Distribution	Normal
Method	Exact
Null Mean	103
Mean	100
Standard Deviation	10
Total Sample Size	35
Number of Sides	2
Alpha	0.05

Computed Power
Power
0.407

From our output above, the probability of correctly rejecting the null hypothesis is 0.407.

(b) Under the same conditions as the previous part, how big a sample size is needed to obtain power of 0.75?

In this case, we just have to substitute the power with 0.75 and the ntotal with a dot:

```
proc power;
    onesamplemeans
    test=t
    mean=100
    nullmean=103
    stddev=10
    ntotal=.
    power=0.75;
```

And the output is as follows:

The POWER Procedure
One-Sample t Test for Mean

Fixed Scenario Elements	
Distribution	Normal
Method	Exact
Null Mean	103
Mean	100
Standard Deviation	10
Nominal Power	0.75
Number of Sides	2
Alpha	0.05

Computed N Total	
Actual Power	N Total
0.755	80

As we can see, we need to obtain a sample size of 80 in order to have power of at least 0.75.

(c) SAS gave you an "actual power" that is slightly from what you asked for. Explain briefly why it did that.

Analogous to what was done in R, SAS appears to round the the sample size up (to the nearest integer) for the minimum amount of power of 0.75. This means that the range of the true sample size to obtain a power of 0.75 is (79, 80), but not actually 80. "actual power" is the power of the sample size being equal to 80.

(d) After further study, researchers found that the population SD is about 12 rather than about 10. Calculate the sample size now required to get a power of 0.75, and explain briefly why the change in results from part (b) is not surprising.

Our code ran is as follows

```
proc power;
    onesamplemeans
    test=t
```

```
mean=100
nullmean=103
stddev=12
ntotal=.
power=0.75;
```

and the output of the code above is as follows:

The POWER Procedure
One-Sample t Test for Mean

Fixed Scenario Elements	
Distribution	Normal
Method	Exact
Null Mean	103
Mean	100
Standard Deviation	12
Nominal Power	0.75
Number of Sides	2
Alpha	0.05

Computed N Total	
Actual Power	N Total
0.750	113

We see that the sample size required is 113. This is not surprising because as standard deviation goes up, there is a larger likelihood of incorrectly rejecting the null hypothesis (as there is larger variance for your conclusions). To maintain the power of 0.75, the sample size would have to increase, and it did as expected.